

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2015

CHEMISTRY

9189/2

1

- (a) (i) Mass of carbon in CO_2 6,6 g = Carbon in 3 g

$$\Rightarrow \frac{12}{44} \times 6,6$$

$$= 1,82 \text{ g}$$

mark for the answer accept if the figure rounded off is shown [1]

- (ii) Mass of Hydrogen in water = Hydrogen in 3 g

$$\Rightarrow \frac{2}{18} \times 3,6$$

$$= 0,4 \text{ g}$$

mark for the answer accept 1 d.p. [1]

- (iii) Mass of oxygen = $3 - (1,82 + 0,4)$

$$= 0,78 \text{ g}$$

mark for the answer (2 d.p.) accept if mark was awarded for a [1]

- (b) Empirical Formula

Element	C	H	O
mass	1,82	0,4	0,78
moles	$\frac{1,82}{12}$	$\frac{0,4}{1}$	$\frac{0,78}{16}$
	0,15	0,4	0,05
Divide by	$\frac{0,15}{0,15}$	$\frac{0,4}{0,15}$	$\frac{0,05}{0,15}$
smallest	1	2,67	0,33
	3	8	1

ecf.

[1] mark for the working

$$\text{EF} = \text{C}_3\text{H}_8\text{O}$$

$$M_r = 60$$

EF must follow from the calculation [1] mark the E

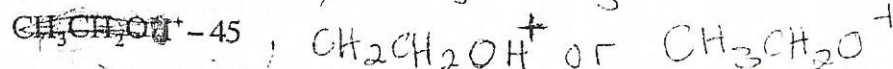
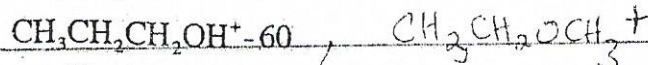
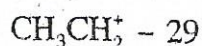
* Award mark for E.F. and M.F. with no markings (BOP)

MF must follow from the answer

$$\therefore \text{Molecular formula} = \text{C}_3\text{H}_8\text{O}$$

[1] mark for the mole formulae

- (c)



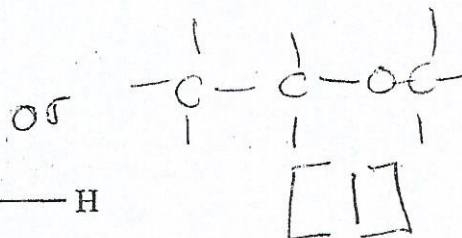
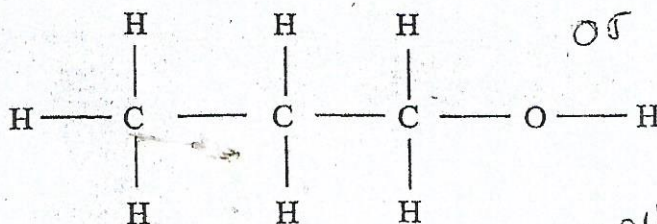
* Penalise if species are not positively charged.

[1] for 2 correct

[2] for all four correct.

[3] for all four correct.

(d)



2

(a)

(i)

$$K_p = \frac{P^2 \text{NO}_2}{P \text{N}_2\text{O}_4}$$

units = atm/Pa | Nm⁻², kPa

(ii)



$$\text{No of moles N}_2\text{O}_4 = \frac{10}{100} \times 4$$

$$= 0.4 \text{ mol}$$

[1] mark for answer

Wrong working correct answer

(R)

Since reaction 1:2

$$\text{No of moles NO}_2 = (4 - 0.4) \times 2$$

$$= 3.6 \times 2$$

$$= 7.2 \text{ mol}$$

[1] mark for answer

$$(iii) \quad K_c = \frac{(7.2)^2}{0.4}$$

$$= 129.6 \text{ mol dm}^{-3}$$

e.c.f. accept rounded up to 130
(units)
wrong units (R)

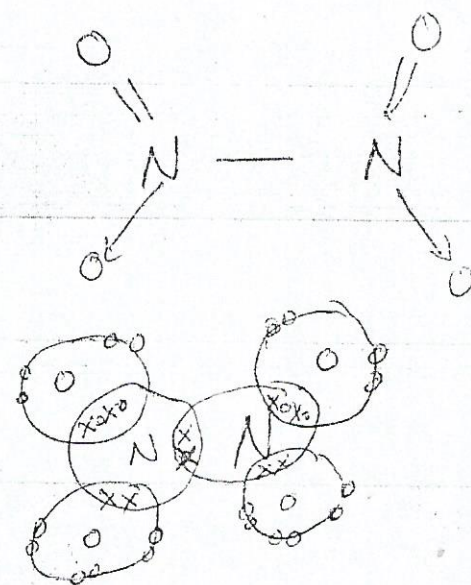
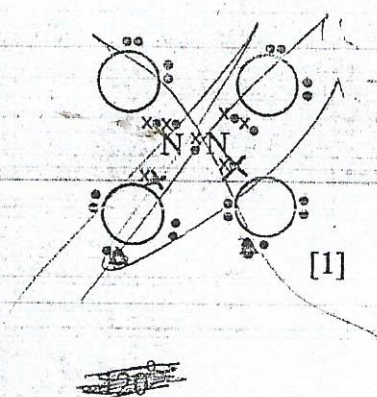
[1]

(iv) K_c value would be lower/ less/ reduced;

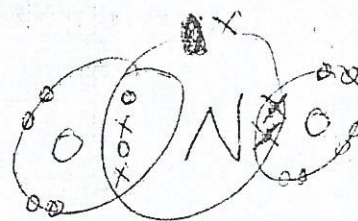
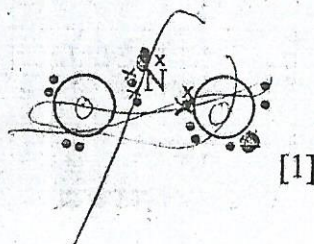
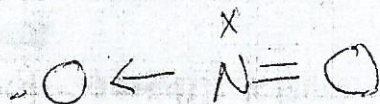
[1]

(Dissociation is endothermic, a decrease in temperature) results in a smaller degree of dissociation hence a smaller value of K_c/ A.W; [1]

(b) (i) 1.



2.

(ii) 120° ;

[1]

3 (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}; [\text{Ar}] 3d^{10};$

[1]

(b) (i) a (d-block) element forming one or more stable ions with partially filled orbitals;

[1]

(ii) is one that contains a central metal ion or a transition metal ion bonded to other atoms, ions or molecules (called ligands);

[1]

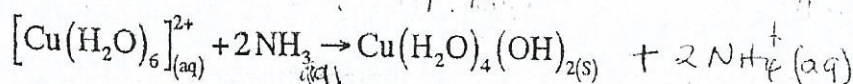
(c) (i) Observations: blue precipitate;
to a dark blue solution in excess

[1]

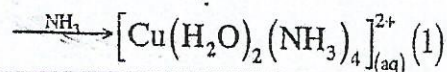
[1]

explanation: ligand exchange

[1]



blue ppt



deep blue solution

please use state symbols.

(ii) - increases; [1]

- atomic mass increases [1]

- atomic radius invariant; *decreases slightly* both points give one mark. [1]

(d) - stronger metallic bond; [1]

- presence of 3d electrons in addition to 4s which participate in metallic bonding / AW more e⁻ involved in metallic bonding [1]

4 (a) (i) 1 mole alcohol produces 0.5 mole of gas [1]

1 mole of gas = 22.4 dm³ / 24 dm³

0.5 moles = 11.2 dm³ / 12 dm³

~~3 moles of gas (0.025) = 0.56 dm³~~

$\frac{0.56}{11.2/12} \times 0.5$

= 0.025 moles

~~11.2 x 3.7 = 41.44~~
~~0.56~~

$\frac{p}{p^\circ} n(\text{alcohol}) = 0.05 \text{ moles}$

(ii)

$M_r = \frac{3.70}{0.05} = 74$

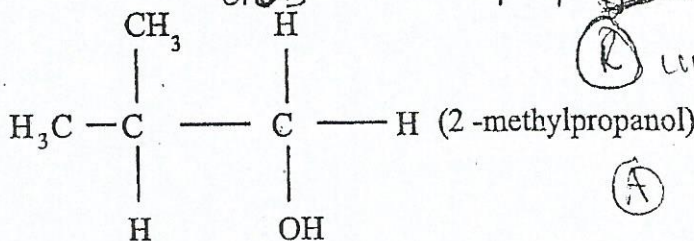
74

$\frac{74}{12} = 6.16$

or 79.28

with 24

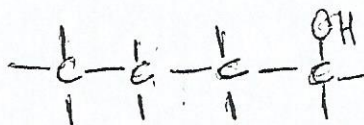
[1]



2 units

mark for the answer

(A) 1° alcohol consistent with M_r obtained.



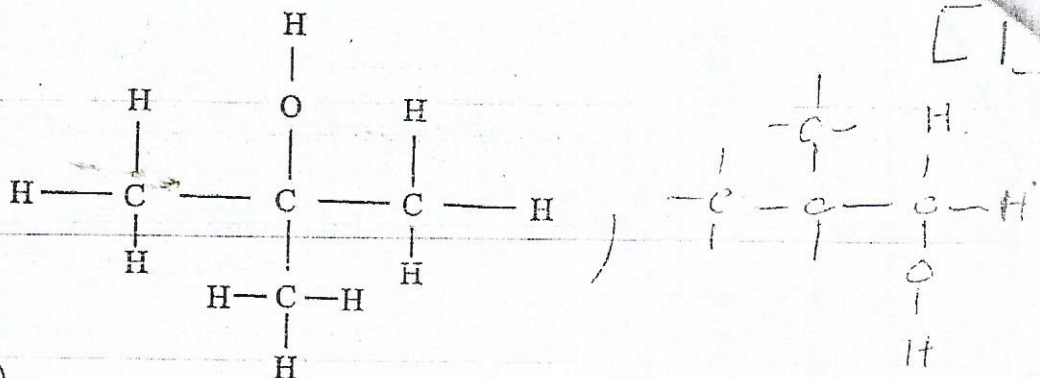
[1]

an aldehyde can be reduced to a 1° alcohol
(a primary alcohol) can be oxidised to carboxylic acid; [1]

forms an alkene on reaction with H₂SO₄; [1]

(b) (i) positional isomerism [1]

(ii)

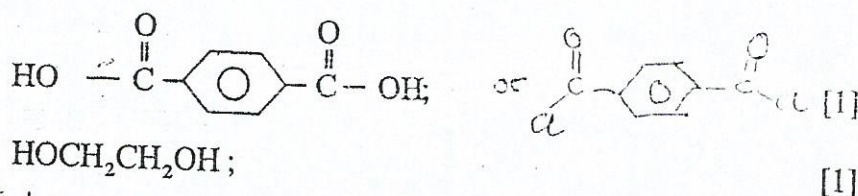


- (warm)
 (iii) Δ add aqueous alkaline iodine / *iodoform test*,
 $\text{NaOH} / \text{I}_2(\text{aq})$ and $\text{HCl}(\text{aq}) \rightarrow$ immediate cloudiness with ~~Z~~ and not
 - Add $\text{ZnCl}_2(\text{aq})$ and $\text{HCl}(\text{aq}) \rightarrow$ immediate cloudiness with ~~Z~~ and not
 - Oxidize with dilute $\text{K}_2\text{Cr}_2\text{O}_7$ with ~~Z~~
 Z gives a yellow ppt;
 (A) alternative methods that are correct.
 W does not change

[1]

5

(a) (i)



[1]

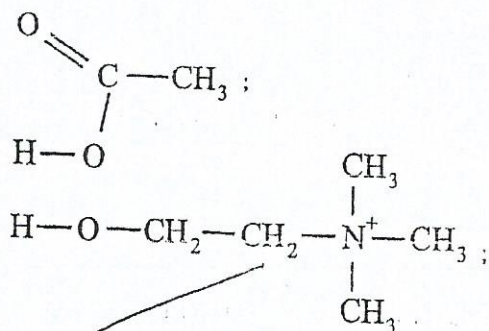
(ii) 1 Heat with acid catalyst

~~2~~ condensation (polymerisation)

warm, boil
 (a) room temp with acyl chlor.
 (R) reflux

[1]

(b) (i)



[1]

[1]

(11)

cross out

(c) (i)

a white ppt is formed;

[1]

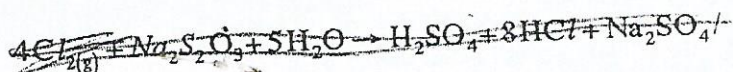
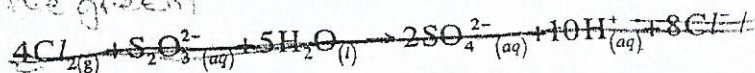


[1]

(ii) the salt dissolved;

[1]

- (iii) Chlorine is decolourised/ colour of chlorine changes from yellowish green to colourless;



{H}

[Total:12]

- 6 (a) (i) 1. hydroxyl, (R) alcohol [1]

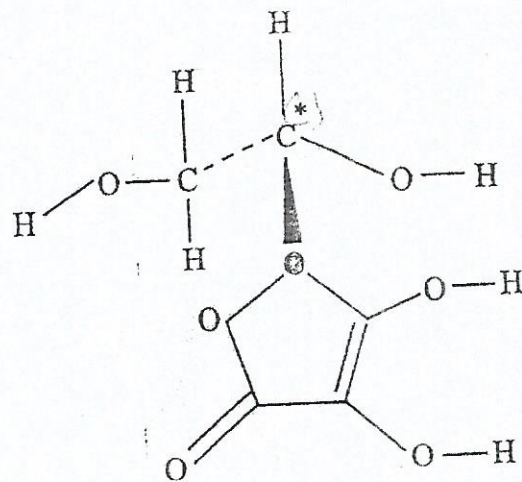
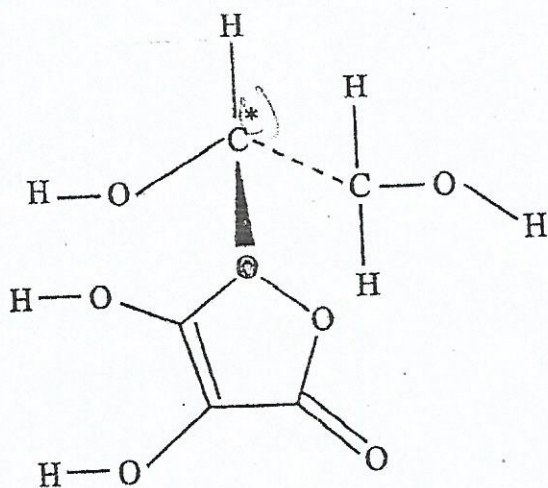
ester/ [1]

carbon to carbon double bond; (R) alkene.

(ii) homologous series for functional groups. (R) Formula; (4) any two correct.

optical isomerism; [1]

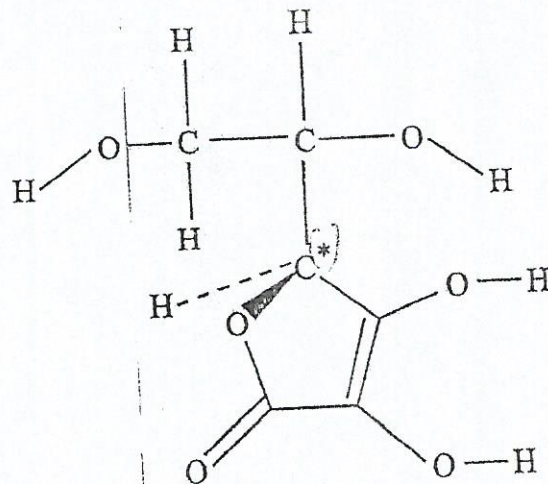
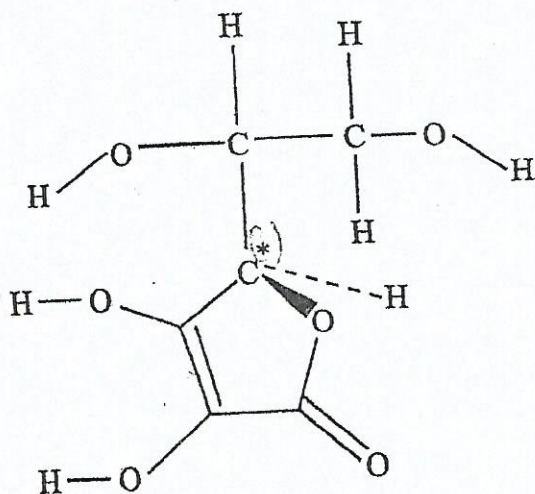
(ii)



mirror line

2 bonds optical isom
1 mirror line and

(CR)



mirror line

3-D structure and
accepted.

Ascorbic acid forms H-bonds with water

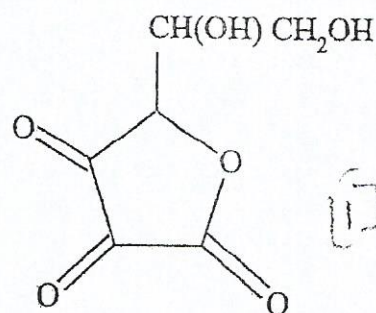
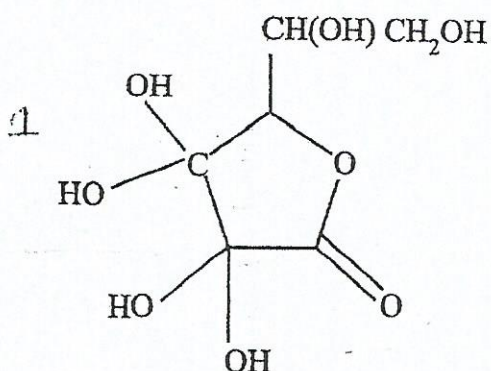
(iii) OH groups can form hydrogen bonds with water molecules; [1]

(b) (i) 1. KMnO_4 changes colour from purple to green/ brown/ is ~~decolourised~~; [1]

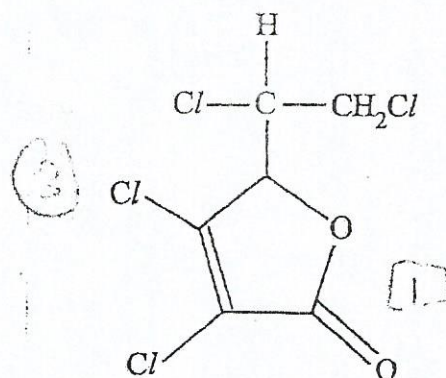
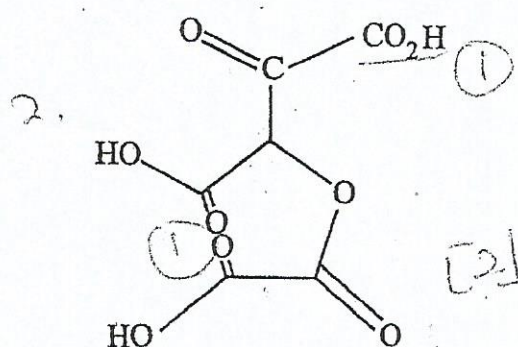
2. decolourisation of KMnO_4 / change colour to pale pink; [1]

3. white fumes (of $\text{HCl}_{(g)}$); [1]

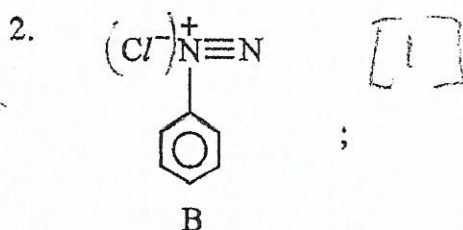
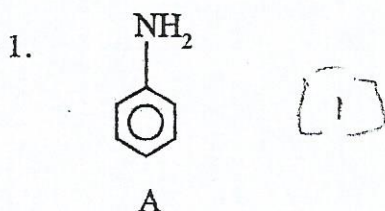
(ii)



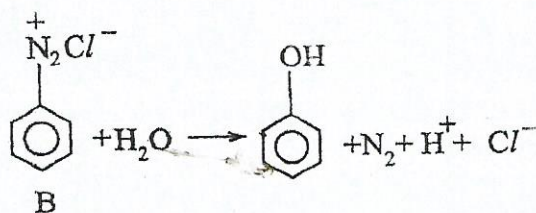
ignore displayed formula.



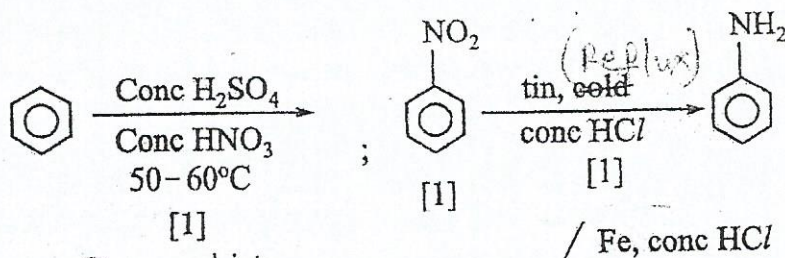
7 (a) (i)



(ii) (iii)



(iii) (b) (i)



Everything must be correct

/ Fe, conc HCl

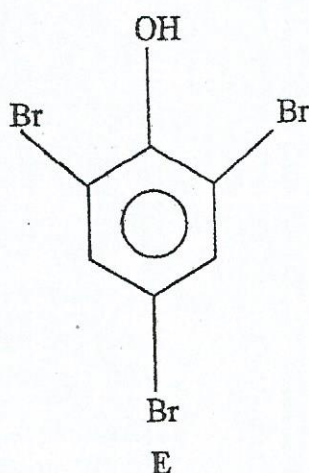
/ LiAlH₄ in dry-ether / NaBH₄ in methanol/ H₂, Ni cat, 140-150°C - 200°C

(ii) Coupling reaction;

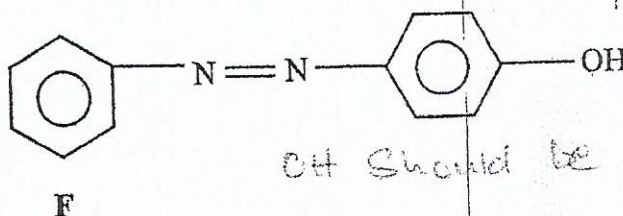
[1]

b(i)

①



2.



accept substitution on position 2.

OH should be on position 4

b(ii) Coupling reaction / Electrophilic substitution

cold NaOH_(aq);/ temperature / 10 °C, NaOH_(aq)

[1]

(iii)

white precipitate;
bromine decolourised;

[2]

(ii) white fumes of HBr.

[Total: 12]

8

(a) (i)

electrophilic addition,
condensation [1] polymerization occurs through
addition-elimination. [1]

A single product is formed in addition polymerisation whereas
condensation polymerisation is accompanied by elimination of
a small molecule such as H_2O / HCl ; [1]

(R) one monomer
for addition
polymerization.

the reactive group is the $C=C$ bond;

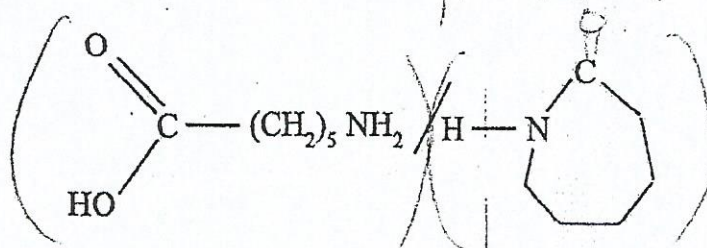
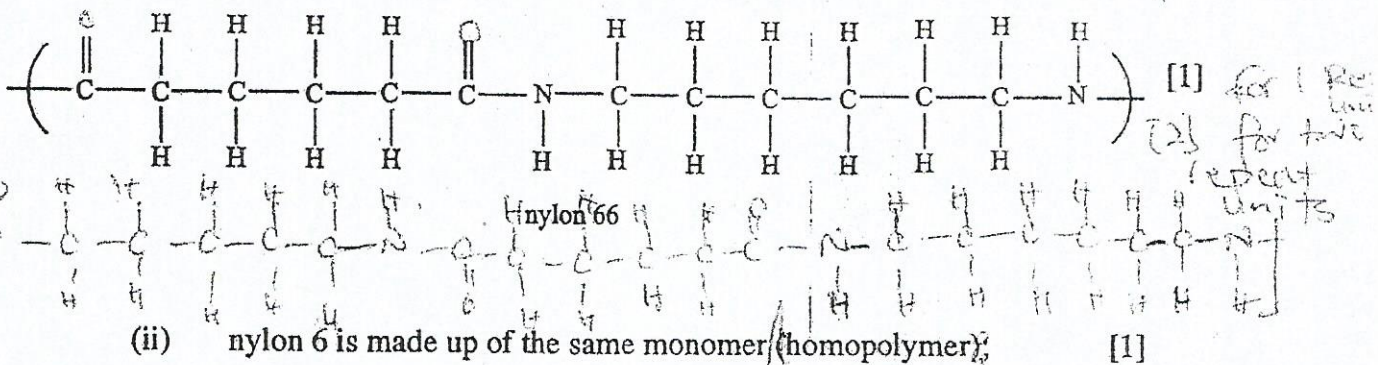
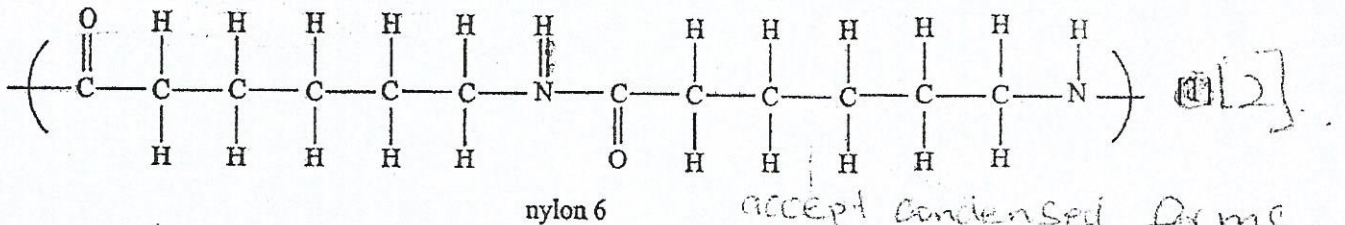
Empirical formula of the monomer is the same as that of the
polymer in addition polymerisation but not in condensation
polymerisation.

A $C=C$ group must be present in the monomer whereas there
are different functional groups in monomers of condensation
polymers.

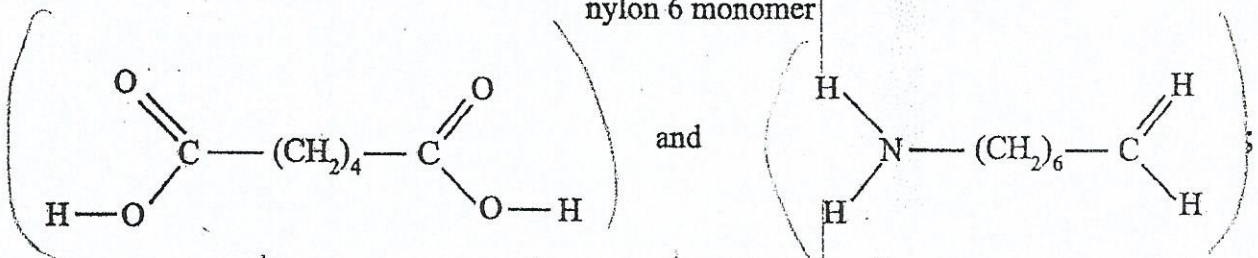
comparison of
the same
property for
the two types
of polymerisation

[any two]

(b) (i)



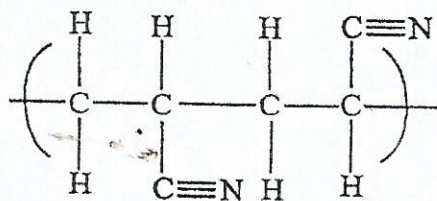
nylon 6 monomer



It's a copolymer made from 2 different
monomers.

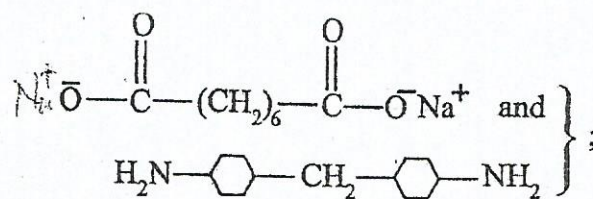
(i) amide (2) peptide; [1]

(ii)



(iii) Polyacrylonitrile does not react with aqueous sodium hydroxide. [1]

quiana undergoes hydrolysis to form [1]



[1]

[1]